EXT GROUPS COMMUTE WITH FLAT BASE CHANGE OUT OF A PSEUDO-COHERENT MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Flat base change for Ext groups. Algebra, Lemma \ref{algebra-lemma-flat-base-change-ext}, cited at proof lines 16723-16724. Rewrites the Ext groups over the target as Ext groups over the source.
- Ext out of a pseudo-coherent module as the base change of the original Ext groups in low degrees. Lemma \ref{lemma-pseudo-coherence-and-ext}, cited at proof lines 16725-16727. Gives the comparison in the stated range of degrees.
- Finite modules over a Noetherian ring are pseudo-coherent, with the corresponding Ext range. Lemma \ref{lemma-Noetherian-pseudo-coherent}, cited at proof lines 16727-16729. Extends the statement to all degrees in the Noetherian case.

Source reference. Original lemma 087R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10436–10436. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 087N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10433–10433. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 087Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10435–10435. Navigation pointer to a statement the

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Source reference. Original lemma 066E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7933–7933. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to R'$ be a flat ring map. Let M, N be R-modules.*

(1) *If M is a finitely presented R-module, then $\mathrm{Hom}_R(M, N) \otimes_R R' = \mathrm{Hom}_{R'}(M \otimes_R R', N \otimes_R R')$.*
(2) *If M is $(-m)$-pseudo-coherent, then $\mathrm{Ext}^i_R(M, N) \otimes_R R' = \mathrm{Ext}^i_{R'}(M \otimes_R R', N \otimes_R R')$ for $i < m$.*

In particular if R is Noetherian and M is a finite module this holds for all i.

Proof. By Algebra, Lemma 087N we have $\mathrm{Ext}^i_{R'}(M \otimes_R R', N \otimes_R R') = \mathrm{Ext}^i_R(M, N \otimes_R R')$. Combined with Lemma 087Q we conclude (1) and (2) holds. The final statement follows from this and Lemma 066E. □